

Tube Lid Detection and Orientation Estimation

Engineering Design Report

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1 Problem Statement

The objective of this project was to build a perception pipeline capable of detecting tube lids and estimating their orientation under different background and lighting conditions.

Initially, the problem looked relatively straightforward because the lids themselves were approximately circular. However, once experimentation began, the actual challenge turned out to be very different.

The main issue was that many tube lids did not have clearly visible boundaries. In several images, the background intensity and the lid intensity appeared extremely similar, especially under bright lighting conditions. In such cases, even though the object was visually understandable to a human observer, the edge information available to the algorithm became very weak.

Another issue was that the dataset itself was quite small, containing only around seventy RGB images. Because of this, the project intentionally avoided deep learning approaches and instead focused on designing a deterministic classical computer vision pipeline.

As experimentation progressed, the approach gradually evolved from trying to perfectly segment the entire object into trying to reliably estimate geometric structure and local appearance asymmetry even when the object boundaries themselves were incomplete.

2 Understanding the Two Core Subproblems

During the initial stages of experimentation, it became clear that the perception problem could not be treated as a single monolithic vision task.

The complete problem naturally separated into two distinct subproblems:

1. Detection / Localization
2. Structural Understanding / Orientation Estimation

Initially, both were assumed to be tightly coupled because the early contour-based pipeline attempted to solve them simultaneously.

The underlying assumption was:

If the object boundary could be segmented correctly, then the object centre, shape, and orientation would naturally follow.

However, practical experimentation showed that these two tasks behave very differently under real-world imaging conditions.

2.1 Detection Problem

The first subproblem involved determining whether a lid exists at a particular spatial location.

This mainly required:

- stable centre estimation,
- approximate radius estimation,
- robustness to partial edge visibility,
- and tolerance to weak boundaries.

Importantly, the detector did not necessarily require exact object boundaries. Even partial gradient evidence could still support reliable localization.

This observation later motivated the transition toward Hough-circle-based geometric voting methods.

2.2 Segmentation / Structural Understanding Problem

The second subproblem was fundamentally different.

Even after localizing a lid, the system still needed to infer:

- where the tab exists,
- where the joint lies,
- and therefore the orientation of the lid.

This stage depended not only on localization but also on understanding local structural asymmetry around the lid.

Initially, contour geometry was expected to provide this information directly. However, once contour extraction became unreliable under difficult lighting and texture conditions, the orientation stage also became unstable.

This revealed an important conceptual distinction:

Accurate localization does not necessarily imply accurate structural understanding.

A detector can successfully estimate the lid centre while still failing to infer orientation.

2.3 Architectural Consequence

This distinction significantly influenced the final pipeline design.

Instead of attempting to solve both problems using a single representation, the final architecture separated them:

- detection became geometry-driven,
- orientation estimation became appearance-driven.

The final detector focused only on robust localization using geometric evidence, while the orientation module independently analysed angular appearance asymmetry around the detected lid region.

This separation made the system substantially more stable because each stage solved only the specific problem it was best suited for.

3 Initial Detection Pipeline

The earliest implementation relied on contour-based segmentation.

The initial pipeline consisted of:

1. grayscale conversion,
2. Gaussian blur,
3. adaptive thresholding,
4. morphological operations,
5. contour extraction,

6. contour filtering using area and circularity.

The reasoning behind this design was initially strong because:

- adaptive thresholding was expected to handle varying illumination,
- contours naturally provide geometric information,
- circularity filtering aligned with lid geometry,
- and contour protrusions could directly support orientation estimation.

Conceptually, the approach was elegant because both detection and orientation relied on the same contour representation.

4 Failure Analysis of Contour-Based Detection

Despite appearing theoretically reasonable, the contour-based approach failed repeatedly under practical conditions.

4.1 Sensitivity to Background Texture

The dataset contained highly inconsistent backgrounds such as:

- wooden tables,
- textured surfaces,
- dark trays,
- and fabric-like materials.

Although adaptive thresholding locally adjusted intensity thresholds, textured backgrounds themselves generated strong local gradients.

This caused:

- fragmented contours,
- noisy connected components,
- merged regions,
- and unstable object boundaries.

As a result, contours no longer represented reliable lid geometry.

4.2 Weak Boundary Contrast

Another major issue appeared when lids were placed on bright surfaces.

Several lids had extremely weak contrast against the surrounding background. In these regions:

- edge gradients became weak,
- thresholding merged the lid into the background,
- contour closure failed.

This failure mode became especially severe for:

- corner lids,
- partially illuminated lids,

- and lids near textured bright regions.

The contour extraction stage fundamentally depended on stable binary segmentation. Once segmentation failed, every downstream stage also became unstable.

4.3 Orientation Instability

The original orientation strategy attempted to estimate the tab direction using contour protrusions.

The underlying geometric assumption was:

The tab protrudes slightly beyond the circular body of the lid, therefore the farthest contour region should correspond to the tab direction.

A radial contour profile was constructed around the centroid and protrusion peaks were analysed.

However, this failed in practice because contour quality itself became unstable.

Observed issues included:

- noisy protrusions caused by threshold artefacts,
- fragmented boundaries,
- unstable radial peaks,
- and inconsistent orientation vectors.

This revealed that relying too heavily on exact contour geometry made the entire pipeline fragile.

5 Intermediate Experiments

Several alternative approaches were explored before arriving at the final architecture.

5.1 Ellipse-Based Detection

Because some lids appeared slightly distorted under perspective effects, ellipse fitting was explored.

The assumption was that perspective distortions could make ellipse-based geometry more suitable than strict circular fitting.

Both OpenCV ellipse fitting and skimage ellipse Hough transforms were experimented with.

However, several problems emerged.

5.1.1 Computational Complexity

Ellipse Hough transforms introduced significantly larger parameter spaces because ellipse estimation required:

- centre,
- major axis,
- minor axis,
- and rotation angle.

This made the detector computationally expensive and unstable.

5.1.2 False Positives

Textured backgrounds generated numerous edge fragments that accidentally voted for large ellipses.

This produced:

- giant false ellipses,
- unstable detections,
- and excessive parameter sensitivity.

Small parameter changes drastically altered detector behaviour.

5.1.3 Geometry Mismatch

Although perspective distortions existed, the lids were still fundamentally much closer to circles than arbitrary ellipses.

The ellipse parameter space therefore became unnecessarily flexible.

This reduced robustness rather than improving it.

The ellipse approach was eventually abandoned.

6 Transition Toward Hough Circle Detection

At this stage, an important conceptual shift occurred.

Initially, the project attempted to recover exact object boundaries.

However, repeated experimentation showed that exact segmentation was not strictly necessary for robust localization.

A key observation was:

Even when contour extraction failed, circular edge evidence still remained visible.

This suggested that a voting-based geometric method could still recover stable localization hypotheses even under partial boundary visibility.

The problem was therefore reframed as:

- robustly estimate centre and radius,
- rather than perfectly recover object boundaries.

This became one of the most important architectural transitions in the project.

7 Final Detection Architecture

The final detector relied on:

- LAB colour preprocessing,
- CLAHE enhancement,
- median filtering,
- Hough circle detection,
- appearance-based filtering,
- and spatial outlier rejection.

7.1 LAB Color Space

Instead of direct grayscale processing, the image was converted into LAB colour space. The L channel isolates brightness information while reducing sensitivity to colour variations.

This improved robustness because several lids had grayscale intensities extremely similar to the surrounding background.

LAB preprocessing improved local brightness representation without strongly amplifying irrelevant colour textures.

7.2 CLAHE

Contrast Limited Adaptive Histogram Equalization (CLAHE) was introduced on the L channel.

This became necessary because weak lid boundaries still remained difficult to distinguish under inconsistent lighting.

CLAHE improved:

- local contrast,
- weak edge visibility,
- and boundary separation.

Importantly, CLAHE enhanced local contrast rather than globally stretching the histogram.

7.3 Median Blur

After CLAHE enhancement, background texture noise became stronger.

Initially Gaussian blur was used, but it softened weak circular edges excessively.

Median blur produced better results because it:

- preserved edge structure,
- reduced texture artefacts,
- and maintained circular boundary sharpness.

7.4 Hough Circle Detection

The final detector used constrained Hough circle estimation.

Several parameters required careful tuning:

- minimum radius,
- maximum radius,
- accumulator thresholds,
- edge thresholds,
- and minimum centre distance.

The radius range was intentionally constrained because all lids occupied a narrow scale range.

This reduced:

- false positives,
- unstable scale estimation,
- and accidental background matches.

8 Appearance-Based Filtering

Although Hough circles improved localization significantly, another issue emerged:

Empty rack holes and background structures were also circular.

Pure geometry alone was therefore insufficient.

To address this, an appearance-based filtering stage was introduced.

For each detected circle:

1. a circular mask was created,
2. mean intensity inside the region was computed,
3. dark regions were rejected.

This worked because actual lids exhibited different brightness characteristics than empty rack holes.

9 Spatial Outlier Rejection

Even after appearance filtering, occasional false detections still remained.

These generally appeared:

- isolated from the tray,
- near unrelated circular structures,
- or around textured regions.

9.1 Attempted DBSCAN Clustering

DBSCAN clustering was initially explored because the lids naturally formed spatial groups.

However, DBSCAN performed poorly because:

- each image contained only around 3–6 detections,
- the spatial distribution was sparse,
- density assumptions became unstable.

Small parameter changes caused valid lids to disappear or false detections to remain.

The dataset was simply too sparse for robust density-based clustering.

9.2 Final Outlier Rejection Strategy

Instead of clustering, a simpler statistical strategy was adopted.

The final method computed:

1. centroid of detections,
2. distance of each detection from centroid,

3. mean and standard deviation of distances.

Detections too far from the normal spread were rejected.

The threshold became:

$$\text{threshold} = \mu_d + k \cdot \sigma_d \quad (1)$$

where μ_d is the mean distance, σ_d is the standard deviation of distances, and k is a tunable multiplier, rather than a hardcoded constant.

This worked better because:

- true lids formed compact spatial groups,
- false detections were usually isolated,
- and the threshold adapted automatically to tray spread.

10 Final Orientation Estimation Strategy

The orientation module underwent the largest conceptual redesign.

Initially, orientation estimation relied on contour protrusions. However, once the detector transitioned toward Hough circles, contours no longer contained meaningful structural information.

Synthetic circular contours generated from Hough detections contained no actual tab geometry.

This forced a complete redesign.

10.1 Key Insight

Although contour geometry became unreliable, the tab still produced:

- local brightness asymmetry,
- subtle appearance protrusions,
- and directional intensity variation.

Therefore, orientation estimation was redesigned to rely on image appearance rather than contour geometry.

10.2 Angular Intensity Profile

The final method constructed an angular intensity profile around the detected lid.

The process consisted of:

1. dividing the circle into angular bins,
2. sampling pixels slightly outside the detected radius,
3. measuring local intensities,
4. constructing an angular profile,
5. smoothing the profile using Gaussian filtering,
6. and detecting dominant peaks.

Sampling slightly outside the fitted circle was important because the tab extends beyond the nominal circular boundary.

This became one of the most important conceptual improvements in the orientation pipeline.

10.3 Peak Selection Logic

Simple maximum selection proved unstable because textured backgrounds generated many local maxima.

To improve robustness:

- peak prominence was analysed,
- local peak width was estimated,
- and a combined score was computed.

The strongest structurally meaningful peak was then selected.

This reduced sensitivity to:

- noise spikes,
- texture artefacts,
- and minor local intensity fluctuations.

11 Pipeline Modularization

As the project evolved, the codebase was modularized into separate components.

Module	Responsibility
<code>detector.py</code>	Lid localization
<code>orientation.py</code>	Orientation estimation
<code>pipeline.py</code>	Pipeline orchestration
<code>matching.py</code>	Prediction-to-GT assignment
<code>metrics.py</code>	Evaluation metrics
<code>visualization.py</code>	Debug overlays
<code>evaluate.py</code>	Benchmarking and robustness evaluation

Table 1: Pipeline modules and their responsibilities.

This separation significantly improved:

- debugging speed,
- readability,
- experimentation,
- and maintainability.

12 Robustness Evaluation Philosophy

Because the dataset was small, the project deliberately avoided pretending that large-scale training was feasible.

Instead, the focus shifted toward robustness evaluation.

The evaluation stage introduced perturbations such as:

- brightness changes,
- Gaussian noise,
- and blur.

The objective was not synthetic data expansion, but rather controlled stress testing of the perception pipeline.

This aligned naturally with industrial deployment philosophy where robustness matters more than benchmark overfitting.

13 Final Design Philosophy

The final pipeline became fundamentally different from the original contour-based system.

The architecture evolved from:

exact boundary extraction

into:

robust geometric localization combined with appearance-based reasoning.

The final system intentionally prioritises:

- robustness over geometric perfection,
- stability over fragile precision,
- interpretability over black-box learning,
- and modularity over monolithic design.

Most importantly, the final architecture was shaped through repeated observation of real failure cases rather than fixed theoretical assumptions.

14 Key Lessons Learned

1. Perfect segmentation is often unnecessary for robust localization.
2. Contour extraction becomes fragile under weak contrast and textured backgrounds.
3. Voting-based geometry can outperform exact segmentation under imperfect conditions.
4. Appearance cues become essential when geometry alone becomes ambiguous.
5. Sparse industrial layouts do not always suit clustering approaches like DBSCAN.
6. Orientation estimation can rely on appearance asymmetry rather than exact geometry.
7. Robustness evaluation can be more valuable than dataset scaling for classical computer vision systems.
8. Failure analysis is often more valuable than theoretical elegance.

15 Conclusion

The final perception system became:

- lightweight,
- deterministic,
- geometry-guided,
- appearance-aware,
- modular,
- and robust to moderate perturbations.

Most importantly, the pipeline evolved through iterative observation, redesign, debugging, and refinement rather than fixed assumptions.

The final architecture therefore reflects a practical engineering approach to perception system design rather than a purely textbook implementation.